

WIMU - LIT

Literature Overview: Credibility of Benchmark Evaluation for Audio-Language Models

The main goal of this project is to review the credibility of benchmark evaluation for audio-language models (ALMs). We approach this from three angles: (1) methods for detecting benchmark data contamination in LLMs and VLMs, which focus on the outcomes of contamination (inflated benchmark scores); (2) membership inference attacks (MIAs), which focus on the cause of contamination by attempting to detect overlap between training and benchmark data; and (3) a survey of state-of-the-art audio-language models and their training/evaluation datasets, identifying where training and evaluation data are disclosed and where they overlap.

1. Benchmark Data Contamination Detection Methods for LLMs and VLMs

In assessing the generalization capabilities of multimodal models, a growing body of work investigates whether benchmark data has leaked into the training pipeline, leading to inflated performance metrics. We review three recent papers from CORE A* conferences that propose detection methods for this problem.

Paper	Venue	ArXiv / Paper Link	Method Name	Modality	Code Available
Song et al. (2025)	EMNLP 2025	arxiv.org/abs/2411.03823	MM-Detect	VLM (text+image)	Yes: github.com/MLLM-Data-Contamination
Zawalski et al. (2025)	ICLR 2026 (Poster)	arxiv.org/abs/2510.27055	CoDeC	LLM (text)	Yes: https://docs.nvidia.com/nemo/evaluator
Park et al. (2025)	ICLR 2026	arxiv.org/abs/2511.03774	Multi-Modal Semantic Perturbation	VLM (text+image)	Yes: github.com/jadenpark0/mm-perturbation

Paper	Venue	ArXiv / Paper Link	Method Name	Modality	Code Available

2. Membership Inference Attacks (MIAs)

As a complementary approach to benchmark contamination detection, membership inference attacks aim to detect intersection between training datasets and benchmark data at the instance level. While contamination detection methods focus on the outcome (performance inflation), MIAs focus on the cause: determining whether a specific sample was part of the model's training set.

Paper	Venue	ArXiv / Paper Link	Method	Modality	Code Available	Key Metrics
Yeom et al. (2018)	IEEE CSF 2018	arxiv.org/abs/1709.01604	Perplexity threshold	General ML / LLM	available for later paper which compare to it	Membership inference accuracy, overfitting gap
Shi et al. (2024)	ICLR 2024	arxiv.org/abs/2310.16789	Min-K% Prob	LLM	available for later paper which compare to it	AUC, TPR@low FPR
Zhang et al. (2024)	Preprint	arxiv.org/abs/2404.02936	Min-K% ++	LLM	available for later paper which compare to it	AUC, TPR@low FPR
VL-MIA (2024)	NeurIPS 2024	arxiv.org/abs/2411.02902	Entropy-based MIA for VLMs	VLM (text+image)	Yes: github.com/LIONS-EPEL/VL-MIA	AUC, membership inference accuracy

3. Audio-Language Models and Datasets

To investigate the credibility of benchmark evaluation in the audio domain, we survey state-of-the-art audio-language models that disclose their training and evaluation data, alongside frontier models from major labs whose training data is not publicly available but that report performance on open-source benchmarks.

Open-Source Models with Disclosed Data

The following models are fully open-source (weights, code, and training data composition disclosed), making them suitable candidates for contamination analysis. All three are from NVIDIA's Audio Flamingo line and share a common architecture (encoder + cross-attention/LLaVA + LLM backbone).

Audio Reasoner (Xie et al., 2025) is a large audio language model trained on CoTA, a dataset of 1.2M reasoning-rich samples created via chain-of-thought annotation with closed-source models. It achieves strong results on MMAU-mini, AIR-Bench, and MELD. Training data and code are open.

Audio Flamingo 2 (Ghosh et al., ICML 2025) is a 3B-parameter ALM with a custom CLAP encoder. It is trained on ~10M audio-text pairs using a 3-stage curriculum and achieves SOTA across 20+ benchmarks. It introduces AudioSkills (synthetic reasoning data) and LongAudio (long audio understanding up to 5 min). Trained exclusively on public datasets.

Audio Flamingo 3 (Goel/Ghosh et al., NeurIPS 2025 Spotlight) is a 7B-parameter model built on a custom Whisper encoder (AF-Whisper). It handles speech, sound, and music; supports multi-turn multi-audio chat and on-demand thinking. Trained on ~50M audio-text pairs with a 5-stage curriculum. Achieves new SOTA on 20+ benchmarks, surpassing Gemini and GPT-4o-audio.

Frontier / Closed-Data Models

The following models report benchmark results on the same open-source datasets, but their full training corpora are not publicly disclosed, making it impossible to directly verify the absence of benchmark contamination through data inspection.

- [Qwen/Qwen2-Audio-7B-Instruct](#) (Alibaba): Weights available on HuggingFace, but full training data composition undisclosed.
- [mistralai/Voxtral-Small-24B-2507](#) (Mistral AI): Weights available, training data undisclosed.

Training/Evaluation Dataset Overlap

The table below shows which publicly known datasets are used for both training and evaluation by the open-source models reviewed. This overlap is the primary vector for potential benchmark contamination.

Dataset	HF Link	Original Paper	Audio Reasoner (Train / Eval)	AF2 (Train / Eval)	AF3 (Train / Eval)
AudioCaps	OpenSound/AudioCaps	Kim et al. NAACL 2019	train (49,838) / test (975)	train (49,838) / test (975)	train (49,838) / test (975)
AudioSet	agkphysics/AudioSet	Gemmeke et al. ICASSP 2017	train (18.7k balanced) / test (17.1k eval)	train (18.7k balanced) / test (17.1k eval)	train (18.7k balanced) / test (17.1k eval)
Clotho-v2	CLAPv2/Clotho	Drossos et al. ICASSP 2020	train (3,839) / evaluation (1,045)	train (3,839) / evaluation (1,045)	train (3,839) / evaluation (1,045)
MusicBench	amaai-lab/MusicBench	Melechovsky et al. 2023	train (52,768) / test (400)	---	train (52,768) / test (400)
CoVoST 2 (zh-en)	facebook/covost2	Wang et al. 2020	train / test	---	---
MELD	ajyy/MELD_audio	Poria et al. ACL 2019	train (9,989) / test (2,610)	---	---
MusicQA	mu-llama/MusicQA	Liu et al. 2024	---	---	Pretraining + Finetuning / Evaluation
LibriSpeech	openslr/librispeech_asr	Panayotov et al. ICASSP 2015	---	---	train (clean/other) / test (clean/other)
VoxPopuli	facebook/voxpathuli	Wang et al. ACL 2021	---	---	train / test

Additionally, Audio Reasoner uses a large combined training dataset [zhifeixie/Audio-Reasoner-CoTA](#) (1.2M samples with CoT annotations).

Model Summary Table

Model	Paper / ArXiv	Venue	Params	Weights Available	Code Available	Training / Disclosure
Audio Reasoner	arxiv.org/abs/2503.02318	Preprint (Mar 2025)	7B	Yes (HF)	Yes: github.com/xzf-thu/Audio-Reasoner	Yes (CoT/dataset o

Model	Paper / ArXiv	Venue	Params	Weights Available	Code Available	Training Data Disclosure
Audio Flamingo 2	arxiv.org/abs/2503.03983	ICML 2025	3B (LLM) + 203M (encoder)	Yes (HF: nvidia/audio-flamingo-2-0.5B)	Yes: github.com/NVIDIA/audio-flamingo	Yes (public datasets)
Audio Flamingo 3	arxiv.org/abs/2507.08128	NeurIPS 2025 (Spotlight)	7B (LLM)	Yes (HF)	Yes: github.com/NVIDIA/audio-flamingo	Yes (~50k pairs, open source audio data)
Qwen2-Audio-7B-Instruct	Alibaba technical report	N/A	7B	Yes (HF: Qwen/Qwen2-Audio-7B-Instruct)	Partial	No (training data undisclosed)
Voxtral-Small-24B	Mistral AI blog	N/A	24B	Yes (HF: mistralai/Voxtral-Small-24B-2507)	No	No (training data undisclosed)

Summary

- Contamination is a real and documented problem in multimodal models.** MM-Detect (Song et al.) finds significant contamination in proprietary models on older benchmarks, with contamination sometimes originating in unimodal pretraining. CoDeC (Zawalski et al.) provides lightweight detection for LLMs via ICL. Park et al. show that VLM-specific perturbation methods are needed since text-only approaches fail.
- MIAs provide a complementary lens.** While contamination detection measures outcomes (inflated scores), MIAs (Min-K%, VL-MIA) directly test whether specific samples were in the training set. The VL-MIA work shows this is feasible for multimodal models via entropy analysis.
- Audio-language models present a clear opportunity.** Open models (Audio Reasoner, AF2, AF3) disclose training data that overlaps with evaluation benchmarks (AudioCaps, AudioSet, Clotho are used for both training and testing). Frontier models (Qwen2-Audio, Voxtral) report results on the same benchmarks but do not disclose training data. No existing work has applied contamination detection or MIA methods to audio-language models specifically.